

**Polyhedrons** *Excel Review Center*

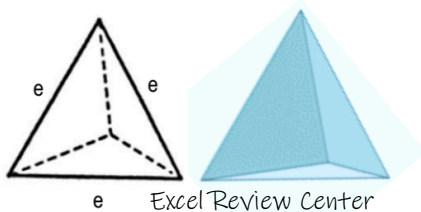
A polyhedron is a solid whose faces are plane polygons. The term “polyhedron” comes from the Greek “poly” for “many” and “hedron” for “base or face”.

Polyhedra are named according to the number of faces. A polyhedron with six faces is called hexahedron while the one with three faces is called tetrahedron.

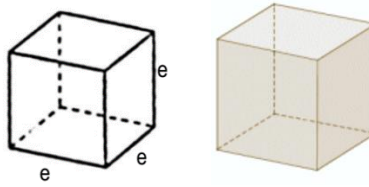
Convex polyhedron is one that lies entirely on one side of a plane that contain any of its faces while a **concave polyhedron** has at least one face so located that there are parts of the polyhedron on both sides of a plane containing that face. If a line that connects any two points on the surface of a polyhedron is completely inside or on the polyhedron, the polyhedron is convex, otherwise it is concave.

Platonic Solids

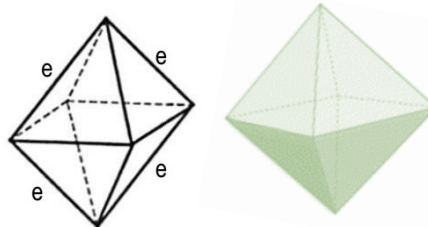
A regular polyhedron is a solid with all its faces identical regular polygons. There are only five regular convex polyhedra, namely the tetrahedron, the hexahedron (cube), the octahedron, the dodecahedron and the icosahedron. These solids are also known as **Platonic solids** in honor of Plato (427 – 348 B.C.).

1. Tetrahedron

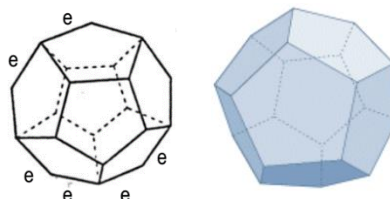
Faces	Triangle
No. of faces	4
No. of edges	6
No. of vertices	4
Polygon angle	60°
Sum of angles	180°
Radius of circumscribed sphere	$\frac{e\sqrt{6}}{12}$
Radius of inscribed sphere	$\frac{e\sqrt{6}}{12}$
Total area	$e^2\sqrt{3}$
Volume	$\frac{1}{12}e^3\sqrt{2}$

*Excel Review Center***2. Hexahedron**

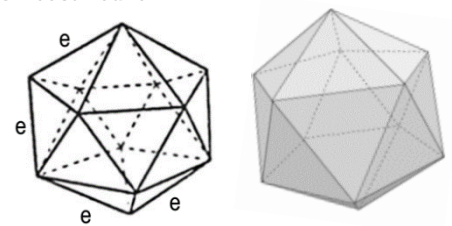
Faces	Square
No. of faces	6
No. of edges	12
No. of vertices	8
Polygon angle	90°
Sum of angles	270°
Radius of circumscribed sphere	$\frac{e\sqrt{3}}{2}$
Radius of inscribed sphere	$\frac{e}{2}$
Total area	$6e^2$
Volume	e^3

3. Octahedron

Faces	Triangle
No. of faces	8
No. of edges	12
No. of vertices	6
Polygon angle	60°
Sum of angles	240°
Radius of circumscribed sphere	$\frac{e\sqrt{2}}{2}$
Radius of inscribed sphere	$\frac{e\sqrt{6}}{6}$
Total area	$2e^2\sqrt{3}$
Volume	$\frac{1}{3}e^3\sqrt{2}$

4. Dodecahedron

Faces	Pentagon
No. of faces	12
No. of edges	30
No. of vertices	20
Polygon angle	108°
Sum of angles	324°
Radius of circumscribed sphere	$\frac{e}{4}\sqrt{3}(1+\sqrt{5})$
Radius of inscribed sphere	$\frac{e}{4}\sqrt{\frac{50+22\sqrt{5}}{5}}$
Total area	$6e^2$
Volume	$\frac{e^3}{4}(15+7\sqrt{5})$

5. Icosahedron*Excel Review Center*

Faces	Triangle
No. of faces	20
No. of edges	30
No. of vertices	12
Polygon angle	60°
Sum of angles	300°
Radius of circumscribed sphere	$\frac{e}{4}\sqrt{2(5+\sqrt{5})}$
Radius of inscribed sphere	$\frac{e}{2}\sqrt{\frac{7+3\sqrt{5}}{6}}$
Total area	$5e^2\sqrt{3}$
Volume	$\frac{5e^3}{12}(3+\sqrt{5})$

Note: Every polyhedron in three-dimensional space consists of (two-dimensional) **faces**, (one-dimensional) **edges** and (zero-dimensional) **vertices**. The faces of the polyhedron intersect each other along the edges, which meet at the corners known as vertices.

Euler's Formula

The Euler formula relates the number of vertices *V*, edges *E* and faces *F* or a polyhedron.

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$$V + F - E = 2$$

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